MIL-T-45919(WC)
13 December 1968
SUPERSEDING
RIAPD-691A
10 April, 1967

MILITARY SPECIFICATION

TARGETS, SILHOUETTE,

STANDING MAN HOLDING PISTOL AND 1000 INCH

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of water-resistant fiberboard targets.

1.2 Classification. Targets shall be of the following types:

Type I - Target, Silhouette, Standing Man Holding Pistol
Type II - Target, Silhouette, 1000-Inch

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

CCC-T-191 - Textile Test Methods

STANDARDS

Military

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

FSC 6920

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DRAWINGS

Army - WC

C8432382 - Target, Silhouette, 1000 Inch
F8432384 - Target, Silhouette, Standing Man Holding Pistol

PUBLICATIONS

Army - WC

P8432382 - Packaging Data Sheet for Target, Silhouette, Type II
P8432384 - Packaging Data Sheet for Target, Silhouette, Type I

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Technical Association of the Pulp and Paper Industry Standards

T411M-44 - Thickness and Density of Paper
T477M-47 - Blocking Resistance of Paper and Flexible Materials
T480M-51 - Specular Gloss of Paper at 75 Degrees
T491SU-63 - Water Immersion Test of Paperboard

(Applications for copies should be addressed to the Technical Association of the Pulp and Paper Industry, 380 Lexington Avenue, New York, New York 12206.)

American Society for Testing and Materials

D747-61T - Method of Testing for Stiffness of Plastic by Means of a Cantilever Beam

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania, 19103.)

3. REQUIREMENTS

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3.1 Material. Targets shall be made from water-resistant fiberboard which shall comply with the requirements of this specification.

3.1.1 Machine direction. Machine direction of the targets shall be lengthwise.

3.2 Dimensions. The targets shall conform to the dimensions and tolerances specified on C8432382 and F8432384.

3.2.1 Thickness. The thickness of the fiberboard targets shall be as specified on the applicable drawings when determined by the test specified in Table II.

3.2.2 Form. The targets shall be furnished cut to the size and shape specified on C8432382 and F8432384.

3.3 Color. Either one side or both sides shall be colored as specified on the applicable drawing.

3.3.1 Colorfastness to light. The colored side(s) of the target shall show a "good" colorfastness to light rating after a minimum of 20 hours exposure to light.

3.3.2 Gloss. The finished targets shall have a maximum gloss value of 20 units.

3.4 Water absorption. The water absorption of the targets shall be a maximum of 10 percent.

3.5 Delamination. Type I targets shall not delaminate at test temperatures of 0° and 75°F.

3.6 Blocking. The targets shall show no evidence of blocking or adhesion.

3.7 Weight. The weight of the Type I target shall be a maximum of 2.75 pounds per target. The weight of the Type II target shall be a maximum of .25 pounds per target.

3.8 Stiffness (modulus of elasticity). The targets shall have a minimum stiffness, in the machine direction, of 184,000 psi after 24 hours of water immersion.

3.9 Puncture resistance. The targets shall have a minimum puncture resistance of 40 pounds after 24 hours of water immersion at 70° - 85°F.

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3.10 Workmanship. The targets shall be free from foreign or undigested matter, discoloration, dirt or oil spots, holes, tears, wrinkles, chafed or scuffed marks. Targets shall be trimmed and shall have no ragged, torn, uneven, or crushed edges.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for lot acceptance.

4.2.1 Inspection lot. All targets manufactured in the same identifiable production period, by the same process under the same operating conditions, and submitted for acceptance by one supplier at one time shall be considered a lot for inspection purposes. A lot size, for examination (see 4.3) and for test (see 4.4.1), shall be expressed in units of boxes. Examination of packaging and packing shall be expressed in units of shipping containers.

4.2.2 Sampling for examination. A random sample shall be taken from each lot in accordance with MIL-STD-105, Inspection Level S-3. The Acceptable Quality Level (AQL) shall be as specified in Table I.

4.2.3 Sampling for test. Sampling for the tests specified in Table II shall be as specified in 4.4.1.

4.2.4 Sampling for preparation for delivery. A random sample of shipping container units shall be taken from each lot in accordance with MIL-STD-105, Inspection Level S-3. The Acceptable Quality Level (AQL) shall be 4.0 percent defective.

4.3 Examination.

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4.3.1 Each sample box of targets taken in accordance with 4.2.2 shall be examined to determine compliance with this specification. Examination shall be conducted in accordance with Table I.

TABLE I - Classification of Defects

Categories	Defects	Method of Inspection
Critical	None defined	
Major	AQL = 2.5 percent defective	
101	Material, incorrect type of material, or not in flat, cut sheets	Visual
102	Color, not as specified, or does not fall within the color limitations	Visual
103	Packaging, packing and marking not as specified	Visual
104	Markings, illegible, incorrect, or incomplete	Visual
Minor	AQL = 4.0 percent defective	
201	Average count, less than specified	Count
202	Workmanship	Visual

4.4 Tests.

4.4.1 Test of targets. Unless otherwise specified in the contract or order, targets shall be tested for the applicable characteristics as indicated in Table II, for each lot presented for examination. The unit of product shall be six (6) targets for the Type I target and twenty-five (25) targets for the Type II target. No more than one sample unit shall be drawn from any one box of targets in the lot. There shall be no evidence of failure to meet the specified requirements.

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TABLE II - Tests

Characteristics	Specification Reference		Requirement Applicable To		Number of Determinations	Results Reported As	
	Para	Test [1] Method	Sample Unit	Lot Avg	Per Sample Unit	Pass or Fail [2]	Number to Nearest
Thickness	3.2.1	T411M	-	X	Avg of 10	-	0.001 in.
Color	3.3	-	-	X	3	X	-
Colorfastness to light	3.3.1	4.4.8	X	-	3	X	-
Gloss	3.3.2	T480M	X	-	Avg of 5	-	0.1 unit
Water absorption	3.4	T491SU [3]	X	-	Avg of 2	-	1.0%
Delamination	3.5	4.4.5	X	-	3	X	-
Blocking	3.6	T477M [4]	X	-	3	X	-
Weight	3.7	4.4.4	-	X	Avg of 3	-	0.05 lb.
Stiffness	3.8	D747 [5] and 4.4.6	-	-	Avg of 5	-	100 p.s.i.
Puncture resistance	3.9	4.4.7	X	-	Avg of 5	-	1.0 lb.

1. Numbers preceded by the letter "T" refer to TAPPI Standard Methods. Numbers preceded by the letter "D" refer to ASTM Standard Methods. Test methods shall be performed in accordance with the applicable procedures, except that the stipulations for the number of specimens to test, number of determinations to perform, or number of results to report shall be as specified in this Table. All testing shall be performed under atmospheric conditions as specified in 4.4.3.

2. If failure is indicated, report either description or numerical point of failure, as applicable.

3. Specimens shall be immersed for 24 hours.

4. Blocking pressure shall be 2 psi, temperature 140°F., and relative humidity 75 percent.

5. A 50-inch-pound Tinins Olsen stiffness tester can be calibrated

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from the Army Metrology and Calibration Center, Frankford Arsenal. The procedure is listed AMCC Calibration Procedure 133.

4.4.2 Test procedures. Each sample of targets, or each sample unit, specified in 4.4.1, and conditioned in accordance with 4.4.3, shall be tested in accordance with the applicable test method specified in Table II and this specification.

4.4.3 Conditioning. Targets shall be conditioned for 48 hours at 73° + 2°F. and 50 ± 4 percent relative humidity prior to testing.

4.4.4 Weight. One target shall be weighed on any balance that is accurate to 0.1 pound. Results shall be expressed in pounds.

4.4.5 Delamination. Type I targets shall show no delamination or separation of plies over 1/8-inch from the edge of specimen when tested as follows:

4.4.5.1 0° Fahrenheit. The bottom four inches shall be cut from the target and conditioned to 0°F. for 2 hours, along with a 12-inch radius mandrel. After conditioning, and at the test temperature, the specimen shall be bent slowly around the mandrel and observed for any signs of delamination or separation of plies.

4.4.5.2 73° Fahrenheit. The specimen, as described above shall be immersed in water at 70° - 85°F. for 24 hours. After removal from the water, the specimen shall be bent slowly around the test mandrel at room temperature (73°F.) and observed for delamination (separation of contiguous layers or plies).

4.4.6 Stiffness (M.E.). Stiffness shall be determined on a suitable testing instrument as described in ASTM D747 with the following exceptions: specimen size shall be 2 x 6-inches, the span length shall be four inches, angular deflection shall be 5° and moment weight shall be 20 in. lbs. The equation for determining stiffness shall be as follows:

$$\text{Stiffness (M.E.) psi} = \frac{4L \times S^3}{D \times W \times T^3}$$

$$\text{Where } L = \frac{\text{Load-scale reading} \times \text{moment (inch pounds)}}{100 \times S}$$

S = Span length (inches)
D = Angular deflection = 2S sin. 1/2 Ø (inches)
W = Width of specimen (inches)
T = Thickness of specimen (inches)

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Test specimens shall be cut to size and measured prior to water immersion. After immersion, excess water shall be blotted carefully from the specimen, exercising care that the specimen is not flexed or compressed.

4.4.7 Puncture resistance. Five 4 x 4 inch specimens shall be immersed for 24 hours in water at 70° - 80°F. After removal from the water, the specimen shall be blotted lightly and tested. Any device capable of applying a known amount of weight to a perpendicularly mounted penetrator shall be satisfactory. The penetrator shall be a 0.30-inch rod, tapered to a sharp point in 3/4 inch. Place the specimen on a wooden or metal block (approx. 3 inches square and 1 inch thick) with a 1/2 to 1 inch diameter hole through the 1 inch thickness. Measure the resistance to puncture by applying the load in 5 pound increments to the penetrator centered on the hole. Complete penetration should take place in less than 5 seconds. If the specimen is not penetrated fully in 5 seconds, move penetrator to another section of the specimen and add an addition 5 pound weight. Record the minimum amount of weight that penetrates the specimen in 5 seconds or less. Repeat the procedure on the remaining specimens.

4.4.8 Colorfastness to light. Method No. 5660 of CCC-T-191 shall be used. Results shall be interpreted as described in Method No. 5662 of the same specification (see 6.5).

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The targets shall be packaged, packed, and marked as specified on Packaging Data Sheets P8432382 and P8432384, as applicable.

5.2 Level. The level of protection shall be as specified by the procuring activity (see 6.2).

6. NOTES

6.1 Intended use. These targets are used for training purposes during either outdoor or indoor target practice.

6.2 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.
(b) Title, number and date of applicable drawing.
(c) Quantity.
(d) Type of target.
(e) Level of protection required.

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6.3 Suggested contractual features.

6.3.1 Color chips. Color chips, as specified on the drawing and in Federal Standard 595, may be obtained from:

Business Service Center
Region 3
Seventh and D Street, S.W.
Washington, D.C. 20024

6.4 Storage. To minimize the effects of natural aging, targets should be stored in a cool, dry location. It is recommended that targets be stored for no longer than 3 years.

6.5 Colorfastness to light. Method No. 5660 of CCC-T-191 is intended for determining the colorfastness of cloth to light. For the purpose of a colorfastness requirement for targets, the use of the word "cloth" in Method 5660 shall be deleted and the words "fiberboard targets" shall be used.

Custodian:

Army - WC

Preparing activity:

Army - WC

Project Number 6920-A040

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either *Government or contractor*, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, *and send to preparing activity.*

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT *CONSIDERED TOO RIGID*

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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